

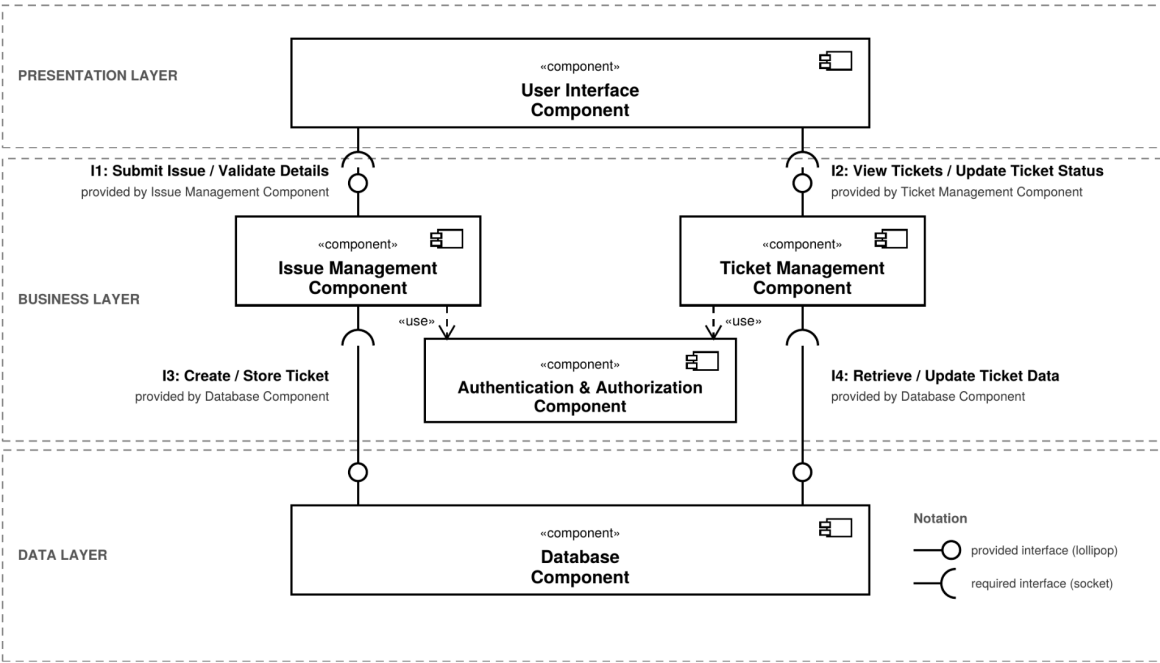
Software Engineering Lab 3

Name: B Nuthan

SRN: PES1UG24CS106

Problem Statement: Hostel Maintenance and Issue Ticketing System

Component Diagram



Architecture Justification

Selected Architecture: Layered Architecture

1. Selected Architecture

The **Layered Architecture** is selected for the Hostel Maintenance & Issue Ticketing System. This architecture divides the system into separate layers, where each layer performs a specific responsibility and communicates with the other layers through defined interfaces.

The system is divided into the following three layers:

2. Presentation Layer

The Presentation Layer contains the **User Interface Component**. It allows hostel residents, maintenance wardens, and maintenance staff to interact with the system. Residents can submit maintenance issues and view ticket statuses, while wardens and maintenance staff can access the functions permitted to their roles.

3. Business Layer

The Business Layer contains the following components:

- **Issue Management Component:** Handles the submission and validation of maintenance issues, including room number, issue category, and description.
- **Ticket Management Component:** Allows wardens to view and assign tickets and allows maintenance staff to update ticket statuses such as *In Progress* and *Resolved*.
- **Authentication & Authorization Component:** Provides role-based access control and ensures that residents, wardens, and maintenance staff can access only their permitted functions.

This layer contains the main processing and business logic of the system.

4. Data Layer

The Data Layer contains the **Database Component**. It stores maintenance ticket details, resident information, issue descriptions, assignments, and ticket status updates. It also retrieves and updates ticket data when requested by the business components.

5. Justification for Selecting Layered Architecture

Layered Architecture is suitable for this system because it provides a clear separation of concerns. Each component has a specific responsibility, making the system easier to understand, develop, test, and maintain.

Changes made to the user interface can be implemented without significantly affecting the database layer. Similarly, changes to ticket-processing logic can be made within the Business Layer without modifying the entire system. This reduces dependency between components and improves maintainability.

The architecture also supports **security** through the Authentication & Authorization Component, which

enforces role-based access. It ensures that different users can access only the functions relevant to their roles.

The separation of layers makes testing easier because each component can be tested independently. The system can also be extended in the future by adding features such as notifications, maintenance history, priority-based tickets, or reporting without redesigning the complete system.

6. Conclusion

Therefore, Layered Architecture is an appropriate choice for the Hostel Maintenance & Issue Ticketing System because it provides **modularity, maintainability, security, easier testing, and flexibility for future enhancements**. The architecture is represented using five components and four interfaces in the UML Component Diagram.